# CS746 Project 1: Scrapy and Data Processing

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# 1 Introduction

The Western Cape is a diverse province with destinations ranging from holiday towns to farming communities to bustling cities. It has become a well-known location for South Africans to move to in the past few years, especially after the Covid-19 pandemic. The ability to work remotely has driven people to areas like coastal towns, with more space, less hustle and a better quality lifestyle [5]. The trend towards moving from metropolitan areas, like Johannesburg, to smaller and larger towns - can also be attributed to a move away from crime towards a safer lifestyle. The fact that municipalities in the Western Cape are some of the better run municipalities in South Africa, makes it even more of an attraction for *semigrants* (when a buyer sells a property and purchase a new one in a new province) [6].

The aim of this study is to answer the question: how has the trend of semigrating to the Western Cape influenced the property market over the past few years? The study is implemented by collecting data from `property24.com` using *Scrapy* - a robust web framework for scraping data from various sources [4]. The main findings of the study are that the semigration trend towards the Western Cape - that started in 2013 [3] - is evident in the increase of the number of properties sold from 2013 to 2016. The study also finds that the Covid-19 pandemic had a large impact on the property market in the Western Cape, and that semigrating to the Western Cape is even more of a trend after the pandemic.

# 2 Implementation

In implementing this study, the general $UR^2IM$ process (figure 1) when working with Scrapy was followed [4]. The **Spider** class in Scrapy defines the custom behaviour for crawling and parsing pages for a particular site (or group of sites) [1], and is therefore responsible for the steps in the $UR^2IM$ process.

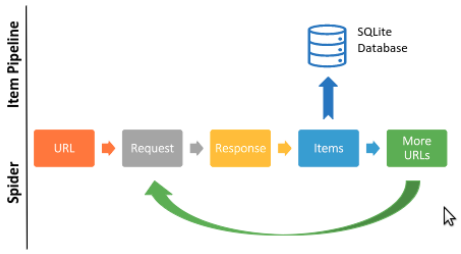


Figure 1: The $UR^2IM$ process [4]

We parse the contents of a page using the built-in Scrapy **Selectors** (XPath and CSS), and generate **Items** (Python objects that define key-value pairs) with the parsed data. As shown in figure 1, the items returned from the spider are persisted to a database. The **Item Pipeline** is responsible for receiving these items, and storing them in the database.

# 3 Crawl process

## 3.1 Items

For our Scrapy implementation, we create two items to store the data scraped from the `property24.com` website:

`Location` Item

1. **name**: the name of the location
2. **type**: province (or city)
3. **num_listed**: the number of properties listed in this location
4. **average_price**: the average price of properties in this location
5. **links**: links to the cities (or suburbs) in this location

`Suburb` Item

1. **name**: the name of the suburb
2. **city_name**: the city in which the suburb is located
3. **street_links**: names of the streets
4. **annual_data**: annual sale and listing trends (raw HTML string)
5. **bybedcount_data**: number of properties listed by bedroom count (raw HTML string)

## 3.2 Property Spider

We start our `property` spider on the '*Property Values in the Western Cape*' page of `property24.com`, with the URL:

$$\texttt{https://www.property24.com/property-values/western-cape/9} \tag{1}$$

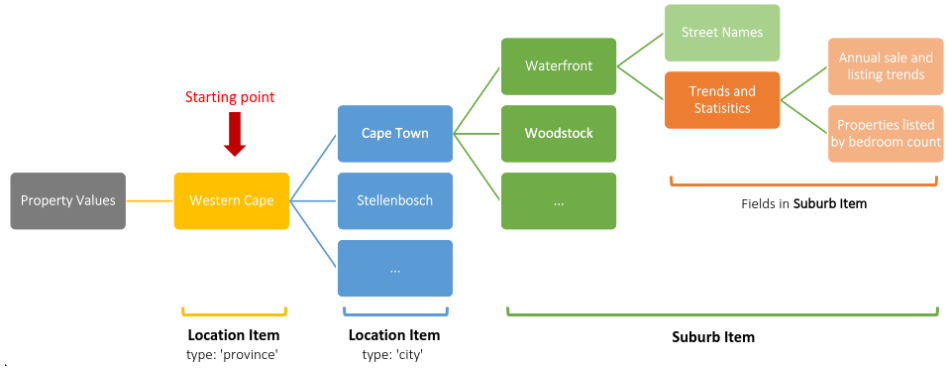


Figure 2: Property Spider

This page contains a list of locations in the Western Cape, where each location is a link to a new page. We call these locations `cities`, and we use CSS selectors to get all the `city` links (href attribute of HTML `<a>` tags) from the page at 1.

For each `city` link, our spider makes a new request. A `city` request bring us to the '*Property values in* {`city`}' page. This page contains a list of areas in that city, where each

area is again a link to a new page. We call these areas `suburbs`, and, again, we use CSS selectors to get all the `suburb` links from the `city` page.

For each `suburb` link, our spider makes a new request. A `suburb` request bring us to the '*Property values in* {`suburb`}' page. This page contains a list of streets in that suburb, as well as a link to the '{`suburb`} *Property Trends and Statistics*' page. We scrape the street names and link using CSS selectors, and make a new request for the scraped link.

This request takes us to the trends and statistics page of a {`suburb`}, which contains - among other things - the annual sale and listing trends for that suburb, and the total number of properties listed by bedroom count in that suburb over the last six months. We scrape this data using CSS selectors.

The trends and statistics page is the last page our spider crawls. Figure 2 shows the crawl process of our `property` spider, where the starting point is the URL given in 1 and every branch from a bright-coloured node after the starting point represents a link that is followed.

### 3.3 Items Pipeline

When our pipeline is instantiated, we setup the connection to our SQLite3 database, and create the tables we need to store items. After all the data required to construct an item has been scraped by our `property` spider, we generate the item, and sent it to the Item Pipeline. In figure 2, the last node covered by an item bracket is the point where we yield that item to the pipeline. When the pipeline receives an item, it uses the Python `isinstance()` function to determine if the item is a `Location` or `Suburb` item, and calls the appropriate handler methods depending on the type. The handler methods cleanse the HTML data, validate the fields in an item, check for duplicates, and store the validated items in the appropriate SQLite3 table. The schema of the SQLite3 database is shown in figure 3.

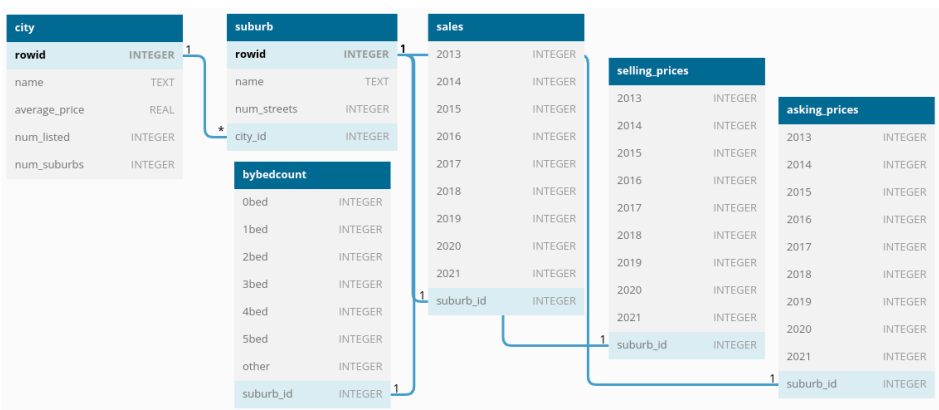


Figure 3: Database Schema

## 4 Data scraped

We scrape the important information on each page we visit, where non-important information includes contact information, advertisements and page navigation links. The aim of the scraping process is to get - for each suburb in all the cities of the Western Cape: the number of sales for 2013 to 2021, the average asking and selling price for 2013 to 2021, and the

number of properties listed by bedroom count for February 2022 to July 2022. This data is scraped from the trends and statistics page of a suburb - the last page visited for each suburb in our crawling process (figure 2).

The annual sale and listing trends are stored in a hard-coded JavaScript element. We use a CSS selector to extract the text within the `<script/>` element, and parse this text with regular expressions to extract the annual number of sales, average selling prices, and average asking prices. We store this data in the `sales`, `selling_prices`, and `asking_prices` tables respectively. The number of properties for sale by bedroom count is stored in a `<table/>` element, and we use CSS selectors to extract the text of each row in the table. The rows are parsed in the Items Pipeline, and the values are stored in the `bybedcount` table.

# 5 Results

The results of this study is divided into three subsections, where each subsection answers a different question using the data gathered during the scraping process.

## 5.1 What is the relationship between the number of properties that are sold and their prices for the top semigration destinations?

According to the article in [6], the top destination for semigrants to the Western Cape is Mossel Bay, followed by the seaside suburb of Milnerton and the bustling city of Cape Town itself. Figure 4 shows the relationship between the number of properties that were sold in these destinations over the last nine years, and the average selling price of properties. We hypothesize that the lower the prices of properties, the higher the sales. We see that this trend is true for properties sold in Mossel Bay (figure 4b), where low average selling prices have high sales, and visa versa. But the opposite seems to hold for Cape Town (figure 4a) and Milnerton (figure 4c), where high average selling prices have high sales. In Milnerton, this relationship seems almost linear. From this results we can deduce that Milnerton and Cape Town are desirable locations for people to move to.

Another interesting thing to note, is the drop in sales from 2019 to 2020, and the increase from 2020 to 2021 - for all three these cities. This result can be explained by the Covid-19 pandemic that happened in 2020. Less properties were bought during 2019 because of the uncertainty caused by the pandemic. After the pandemic, however, more people moved away from metropolitan areas to smaller areas with simpler lifestyles, like Mossel Bay and Milnerton. This is evident in the increase in property sales from 2020 to 2021.

(a) Cape Town (b) Mossel Bay (c) Milnerton

Figure 4: Average selling prices versus the number of sales (2013-2021

## 5.2 How has the trend of semigrating to the Western Cape influenced the average selling and asking price of properties over the past few years?

Figure 5 shows the number of sales in the Western Cape over the last nine years, together with the average selling and asking price of properties (we get these values from the data scraped for each suburb in the Western Cape). It is interesting to note that in 2013, 2015, and 2016, the average selling price was actually higher than the average asking price. This occurrence can be explained by the effect of bidding on selling prices. We see that the number of sales increased rapidly from 2013 to 2016, and reached a maximum in 2016. This coincides with the start of the semigration of buyers to the Western Cape, according to [3]. After 2016 the number of sales decreased slowly, and a large drop in sales is observed from 2019 to 2020. This is most likely due to the Covid-19 pandemic that happened in 2020. The rapid increase from 2019 to 2020 is in alignment with our observation that the Western Cape became a popular destination for semigrants after the pandemic. Also notice that the average asking prices increased moderately from 2013 to 2020, and peaked in 2020 (the same time that sales were at its lowest). The peak in the average asking price in 2020 can be explained by the pandemic, that boosted the demand for housing by increasing the need to work from home, and also the need for more socially distanced housing away from dense urban areas [2].

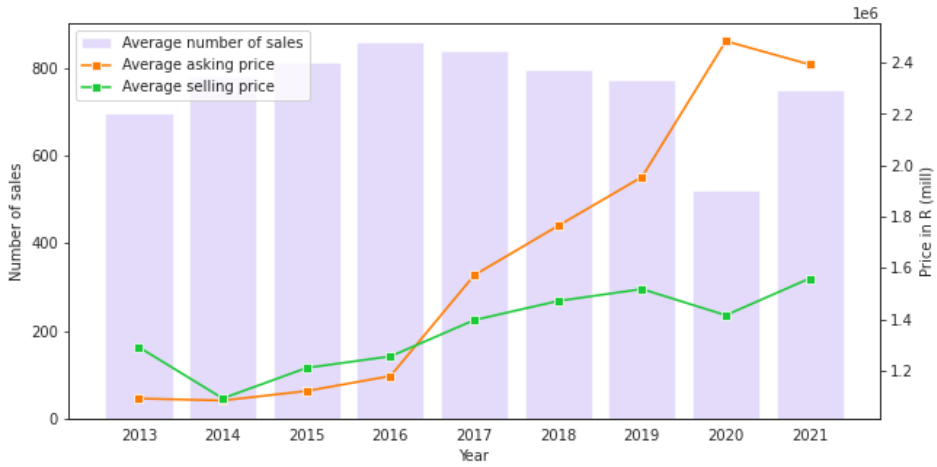


Figure 5: Annual sales and listing trends for the Western Cape

## 5.3 What is the division of the properties listed if categorized by the number of bedrooms? Which bedroom count is seen more on the market for properties listed in the Western Cape?

Figure 6 shows the division of the listed properties by bedroom count over the past six months, for a sample of ten cities in the Western Cape. From the results, it seems that three-bedroom properties are seen on the market the most, followed by two-bedroom properties. *Zero-bedroom* properties (where the living area is not separated from the sleeping area) occurs the least. This trend can be explained by the fact that zero-bedroom properties are usually rented out - meaning that they are used as an investment by owners - and they therefore less

likely to be sold. It is interesting to note that in Stellenbosch, two-bedroom properties are more frequent than three-bedroom properties. This can be explained by the fact that Stellenbosch is well-known for being a student town, and therefore the need for two-bedroom properties are higher.

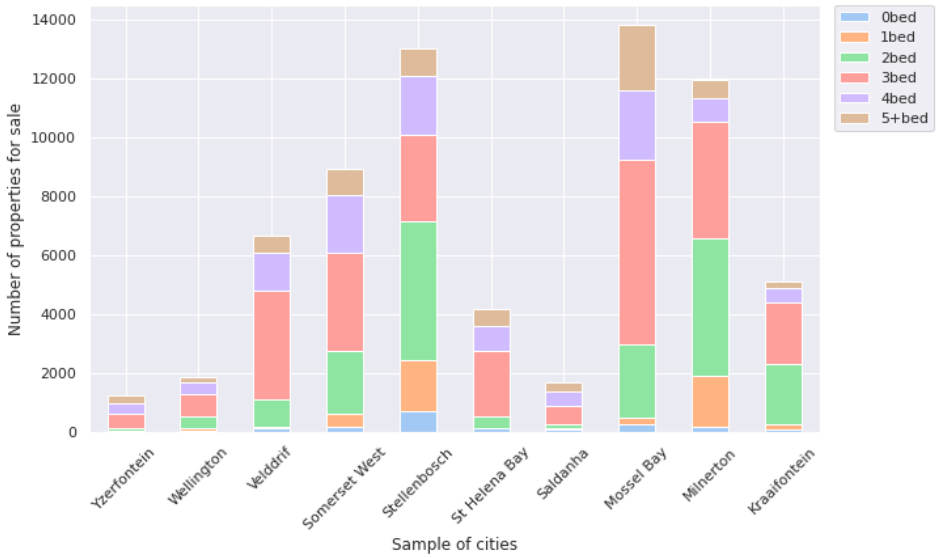


Figure 6: Number of properties for sale by bedroom count for a sample of cities in the Western Cape (Feb. 2022 - Jul. 2022)

Figure 7 shows the division of the listed properties by bedroom count over the last six months, for the entire Western Cape province. We see that three-bedroom properties are listed the most on the market and zero-bedroom properties the least. This is in alignment with the results from figure 6, where a sample of ten cities in the Western Cape was observed.

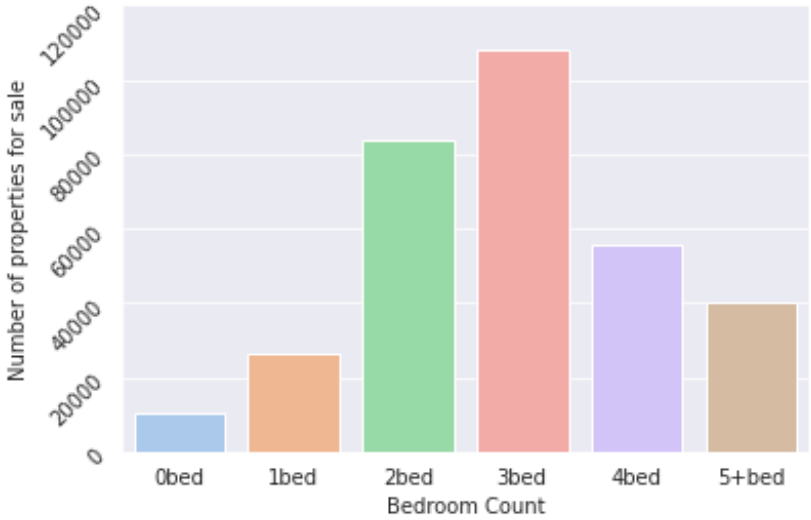


Figure 7: Number of properties for sale by bedroom count for the Western Cape (Feb. 2022 - Jul. 2022)

# 6 Conclusion

This study uses Scrapy to collect data on the values of properties in the Western Cape, from `property24.com`. The study found that the semigrating trend towards the Western Cape has brought about an increase in the number of properties sold. In 2020, the Covid-19 pandemic led to a decrease in property sales because of the uncertainty that people faced. After the pandemic, however, semigration to the Western Cape became even more of a trend as people started moving away from dense, metropolitan areas to simpler lifestyles in areas like coastal towns. This, once again, brought about an increase in the number of properties sold in the Western Cape. The study also investigates Mossel Bay, Milnerton, and Cape Town as the most popular semigration destinations [6]; and found the popularity of Milnerton and Cape Town evident in the high sales of high priced properties. Currently, we can only investigate one year's data after the Covid-19 pandemic. Future work might include researching the long-term affect of Covid-19 on the semigration trend towards the Western Cape, and also to compare this data with the data for Gauteng.

# References


[1] *Scrapy Documentation*. https://docs.scrapy.org/, 2022.

[2] J. V. Duca and A. Murphy. Why house prices surged as the covid-19 pandemic took hold. 2021.

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[4] D. Kouzis-Loukas. *Learning scrapy*. Packt Publishing Ltd, 2016.

[5] G. Williams. The property market: Moving with the times. *finweek*, 2021(13):28–36, 2021.

[6] S. Writer. More gauteng residents are semigrating to the western cape – here's where they are moving to. *BusinessTech*, 8 January 2022.